

Shivam Sahil

Senior AI Platform Engineer | Full-Stack AI Engineer

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SUMMARY

Senior engineer, 6+ years, building production agentic AI systems end to end — multi-agent orchestration on AWS Bedrock AgentCore, MCP tool servers with OAuth 2.0, RAG pipelines, real-time streaming APIs, and the multi-region Terraform infrastructure underneath. Ships from React micro-frontend down to FastAPI, Postgres, and IaC. M.Tech in ML & AI (BITS Pilani). Prior: Apple (Maps), UnitedHealth Group.

SKILLS

AI/Agentic: AWS Bedrock AgentCore, Strands Agents, Google ADK, Model Context Protocol (MCP), multi-agent orchestration, RAG, semantic chunking, vector retrieval, prompt engineering, Langfuse, Claude, Gemini, Llama, Mistral

Languages: Python, TypeScript, JavaScript, Scala, SQL

Backend: FastAPI, NestJS, Node.js, Express, GraphQL, REST, Server-Sent Events, WebSockets, asyncio/asyncpg, microservices, event-driven

Frontend: React 19, Next.js, micro-frontends, Shadow DOM, Zustand, TanStack Query, Redux Toolkit, Tailwind, Vite

Data: PostgreSQL/Aurora, MongoDB, DynamoDB, MySQL, Snowflake, BigQuery, Qdrant, Firebase

Cloud/Infra: AWS (Bedrock, Lambda, ECS, API Gateway, S3, KMS, Cognito, WAF, CloudFront, Route53, VPC), GCP, Terraform, Docker, Kubernetes, CircleCI, GitHub Actions

Security: OAuth 2.0/OIDC, JWT/JWKS, KMS envelope encryption, multi-tenant isolation, RBAC/ABAC, PII detection

Testing: pytest, Vitest, Playwright, MSW, integration & E2E

EXPERIENCE

Software Engineer IV — Mitratesh

Apr 2025 – Present | Remote

Core contributor on Ask Aries, the multi-tenant agentic AI layer for Mitratesh's GRC portfolio — shipping on Alyne today, extending to Prevalent (third-party risk) and Preparis (business continuity). Mitratesh serves 20,000+ enterprise clients across 160+ countries.

- Architected major components of the agentic stack — FastAPI services, AWS Bedrock AgentCore runtime, an MCP tool server, and an embeddable React micro-frontend — deployed multi-region on AWS and designed for rollout across three GRC products.
- Built the real-time streaming spine: resumable Server-Sent Events over async generators and asyncpg pools, delivering live multi-agent reasoning and tool-call events with cursor-based reconnect.
- Delivered an OAuth 2.0 authorization server for the MCP tool server (dual auth, discovery endpoints, AWS WAF), enabling third-party MCP clients to consume enterprise agent tooling securely.
- Led the access-control V2 rewrite: condition-tree policy authorization kernel with registry-backed resource links, migrated from V1 with a consistency gate for zero-downtime cutover.
- Implemented per-tenant envelope encryption (AWS KMS + AES-256-GCM) for all conversational data at rest, with PII detection via AWS Comprehend.
- Owned the multi-region Terraform platform (VPC/peering, Aurora, Elastic Beanstalk, API Gateway, Route53/ACM) with CircleCI OIDC pipelines running migrations on deploy.
- Built the Agent Console UI (React 19, Zustand, TanStack Query) and hardened the embeddable widget with Shadow DOM isolation and streaming that survives view switches.

Senior Software Engineer — ThoughtGenesis (contracted to Apple Inc., Maps)

Nov 2023 – Apr 2025

- Reduced application latency 15% via memoization, caching, and optimized multi-cluster redirection.
- Converted blocking I/O to non-blocking paths; reworked error handling and structured contextual logging.
- Diagnosed gateway timeouts and JVM issues on production clusters via SSH tunnelling and VisualVM remote debugging.

Full Stack Developer — Optum, UnitedHealth Group

Oct 2021 – Nov 2023

- Cut page load time up to 80% by splitting a monolithic GraphQL query into fast and slow paths (DataDog-traced).
- Improved P95 load performance 17% with custom React lazy-loading hooks and bundle optimization.

- Led end-to-end migration of Contentful CMS to Adobe Experience Manager; migrated state from Context API to Redux Toolkit.

Software Engineer — Yoptima Media

Jan 2020 – Oct 2021

- Primary front-end developer on User Management and Campaign Analysis UIs; built Node/Express/NestJS microservices and GCP cloud functions.
- Delivered an end-to-end Next.js campaign dashboard with Chart.js, integrated against Google BigQuery.

SELECTED PROJECTS

Policy Mate — AI Compliance Co-Pilot | *Bedrock AgentCore, Claude, Next.js 15, Terraform, DynamoDB* Supervised multi-agent orchestration for automated GDPR/SOC 2/HIPAA gap detection. Semantic chunking and vector retrieval at >0.85 precision; hash-based dedup and caching cut LLM calls 40%.

Agentic Intelligent Document Processing | *Google ADK, Gemini/Llama/Mistral, Qdrant, FastAPI, React* Multi-agent PDF/DOCX parsing, summarization, and Q&A — 95% synonym recognition, 90% faster processing, 98% chunk reduction via adaptive similarity thresholds and caching.

Micro-Frontend AI Chat Interface | *React 19, Vite, Zustand, Shadow DOM, AWS CloudFront* Dual-mode micro-frontend (widget/container) with Shadow DOM isolation and ES/IIFE bundling for cross-platform embedding; CircleCI CI/CD with Vitest, Playwright, and MSW.

EDUCATION

M.Tech, Machine Learning & Artificial Intelligence (Deep Learning Specialisation, WILP) — BITS Pilani | Nov 2023 – Nov 2025 | CGPA 8.84

B.Tech, Mechanical Engineering (Robotics) — Vellore Institute of Technology | Aug 2016 – Aug 2020 | CGPA 8.87

AWARDS

Second Runner-Up & Most Unique Submission, Grab Hackathon 2022 | Two-time winner, UCW Blockchain Hack | GNEC Award, Mega Hack 2022 | RPA Hackathon Top-7 Finalist | All India Rank 28, Junior Mathematics Olympiad | [more](#)